

What Did Ten Points of Mean Recall Buy? Separating Group-Level from Case-Level Gains in Scene Graph Generation

Anonymous CVPR submission

Paper ID ****

Two frozen models, one cell: relabelling within ϕ removes case identity and nothing else

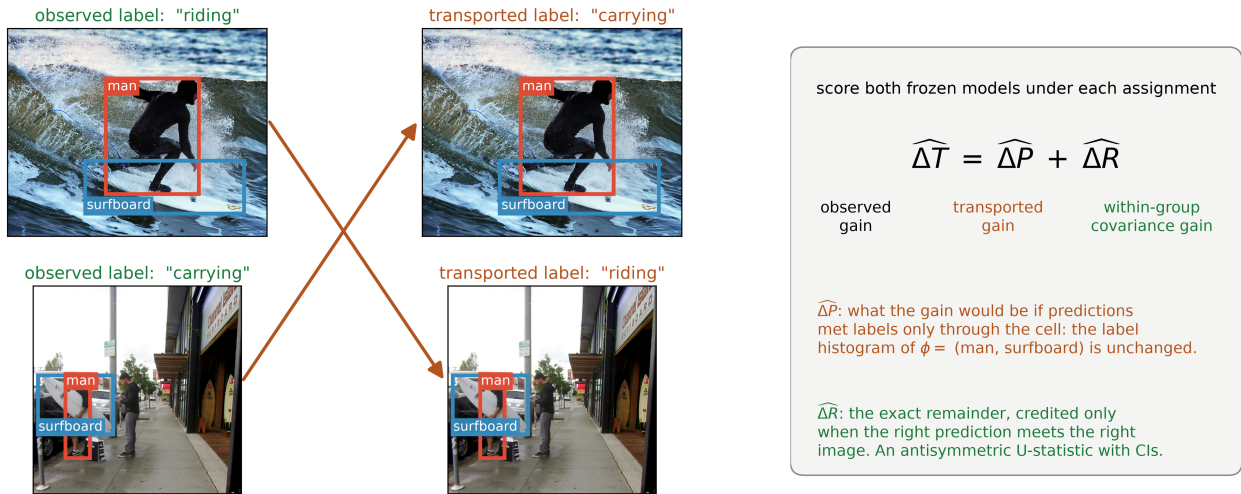


Figure 1. **Within-group relabelling.** Two Visual Genome relations share the subject–object pair (man, surfboard). Swapping their labels keeps the pair’s predicate histogram exactly as it was but breaks the link between each image and its own label. Scoring two frozen models under the observed and the swapped labels splits their score gain exactly: $\hat{\Delta P}$ is the part that survives the swap — earned at the level of the group — and $\hat{\Delta R}$ is the covariance between each relation’s prediction and its own label, which better ranking within the group can supply, and so can a moved operating point.

Abstract

001 Debiasing methods for scene graph generation are judged
002 by mean recall, and one of the best known, Total Direct Ef-
003 fect (TDE), raises it on Visual Genome from 14.6 to 24.8.
004 What did those ten points buy? A model raises mean re-
005 call either by predicting rare predicates more often within
006 each subject–object pair, or by telling which individual re-
007 lations carry them, and the metric credits both alike. Scor-
008 ing each prediction against the label of another relation
009 from the same pair keeps the pair’s predicate frequencies
010 but breaks the link between each relation and its own label.
011 This splits the difference between two frozen models, in a
012 proper score or in mean recall itself, exactly into a group-

level part and a case-level covariance, in $O(NK)$ time with
image-clustered intervals. On TDE’s released checkpoint,
87% of the mean-recall gain (interval 83–91%) is group-
level, and a logit adjustment that changes no ranking within
any pair recovers nine of the ten points at under a tenth of
TDE’s cost in recall. No measure detects a change in TDE’s
ranking within pairs, and its small case-level part changes
sign with the calibration protocol and the label mix. In SG-
Cls the gain is 99% group-level and TDE’s ranking within
pairs gets worse. A second released method, IETrans, gains
21 points, 97% group-level, and loses within-pair discrimi-
nation under every measure.

013
014
015
016
017
018
019
020
021
022
023
024

1. Introduction

Scene graph generation is evaluated on a benchmark whose labels are unusually predictable from something every model sees. Given the ordered object pair (man, surfboard), the predicate on Visual Genome [20] is very often riding, and a lookup table of training frequencies is famously competitive [48]. Recall@ K rewards models that echo this prior, so the field now leads with mean recall, which averages recall over predicates and so rewards getting the rare ones [5, 38]. Debiasing methods are now judged by how far they raise it. Total Direct Effect (TDE) [39], among the most cited of them, raises mean recall@50 of a MOTIFS model from 14.6 to 24.8 on the canonical VG150 split, in our evaluation of the released checkpoint.

What did those ten points buy? A model can move probability towards rare predicates *within each subject-object pair* — say carrying more often for (man, surfboard) — without becoming any better at deciding *which* relations are a man carrying rather than riding. Or it can learn to make that decision. The metric credits both alike. Existing remedies — head/body/tail-stratified recall [23], independent mean recall [25], zero-shot recall [27], rarity-stratified accuracy [19] — slice one model’s performance differently; none attributes the gain *between* two models to one route or the other.

An exact split. Take two frozen models and a test set grouped by a discrete feature both observe, here the subject-object pair (Fig. 1). Score each model’s prediction for one relation against the label of a *different* relation in the same group. This keeps each group’s predicate frequencies, so gain earned at the level of the group survives it, and breaks the link between a relation and its own label, so gain earned on individual cases does not. The observed gain splits exactly into a *group-level* part $\widehat{\Delta P}$ and a *case-level* remainder $\widehat{\Delta R}$, a within-group covariance between the change in the predictions and the label (Theorem 1), for any proper score and for mean recall itself, with no fitted reference model, in $O(NK)$ on cached predictions, with image-clustered intervals.

What it finds. We re-ran TDE’s released checkpoint, re-played the official evaluator exactly so that recall can be computed under any labelling, and split mean recall itself (Sec. 5). Of the ten points, 87% is group-level; the fifteen rarest predicates contribute almost nothing. A thresholded metric also earns case-level credit whenever its operating point moves, so we read the remainder against a control: subtracting the training predicate prior from the baseline’s logits [28], which changes no ranking within any group, recovers nine of the ten points with the same split at under a tenth of TDE’s cost in $R@50$, and against it TDE’s case-

level part is -0.0008 , interval straddling zero. A within-group AUC detects no change in TDE’s ranking, and its proper-score case-level part is small and changes sign with the calibration protocol and the label mix. IETrans [49], which relabels training data within object pairs, gains 21 points, 97% group-level, and loses within-group discrimination under every measure. The converse appears in a controlled study on real pixels (Sec. 6): a model reading frozen CLIP [31] crops, which every aggregate ranks last, carries $+0.00464$ more case-level covariance than the geometry model ranked above it, without ranking better.

Contributions. (i) Audits of two released debiasing checkpoints that split their mean-recall gains exactly, with the mechanism behind TDE’s. (ii) An exact, model-free decomposition of the difference between two frozen models, for every Bregman score and for label-weighted hit rates such as mean recall, with cluster-robust inference, an operating-point control and a threshold-free AUC. (iii) Validation that the case-level part is zero when it should be, covers at the nominal rate under image clustering, and separates calibration, prior and shortcut effects, with protocols for the confounds it does not.

What is not claimed. The scene-graph audits are PredCls, where every model sees the cell, except a replication of TDE in SGCl and SGDet (Sec. 5), where the two models share every predicted object. The split is conditional on the declared grouping: $\widehat{\Delta R}$ measures case-level discrimination *among relations sharing an object pair*, and credits any signal varying inside a pair, including a shortcut nobody recorded. It is not invariant to recalibration — rescaling a model’s confidence moves score between the two parts while changing no ranking — which is why the comparisons whose models differ visibly in confidence, TDE’s and the CLIP model’s, are read confidence-matched. For mean recall the split is of the micro-averaged version, which pools relations across images; it tracks the official per-image average but is not it. And a proper-score case-level part near zero can depend on the calibration protocol and on the label mix; Appendices O and Q report every configuration. These limits are taken up in Sec. 7.

2. Related work

Bias and its evaluation in scene graph generation. Scene graph generation was posed as structured prediction over a whole image, solved by message passing between object and relation nodes [43]; MOTIFS then showed a frequency predictor of the predicate given the object pair to be surprisingly competitive [48]. Mean recall became the headline metric, and debiasing methods now span the causal TDE intervention [39], class-balanced sampling and group-

wise learning [11, 13], predicate-rebalancing message passing [23], predicate-correlation and hierarchical losses [44, 46], prototype embeddings [50], and correction or relabelling of the training labels [22, 49]. Correction by the predicate prior is known to rival them: post hoc, DLFE [6] recovers unbiased predicate probabilities from estimated label frequencies and Structured Sparse R-CNN [41] applies logit adjustment [28]; in training, RTPB [4] resists a prior bias. The metrics themselves have been shown to reward pathologies [25], and open-vocabulary methods inherit them [24]. These works compare methods by their metrics; we ask which part of a metric’s change is group-level. Benchmark audits outside scene graphs find the same pattern: VQA models collapse when answer priors change between train and test [1], GQA-OOD stratifies accuracy by answer rarity within question groups [19], and multimodal models exploit non-visual shortcuts [15, 51]. Gains can also be credited to a mechanism that did not produce them: grounding-based VQA methods keep their out-of-distribution improvement when the grounding signal is irrelevant [35], and methods can exploit how an out-of-distribution split was built [40]. Long-tailed recognition faces the same question for re-weighting and logit adjustment [3, 28]. All of these characterise a *single* model; we account for the gain *between* two.

Proper-score decompositions. Conditional on a grouping, a proper score splits into reliability and resolution terms [9, 29], for general proper scores [2]. Our case-level term is not that resolution term: it descends from the *covariance* decomposition of the Brier score due to Yates [45], which a monotone recalibration moves while the classical resolution is invariant to it. The two coincide only when both models are calibrated within every group. Variance estimators exist for single-model decompositions [36], but not for a contrast between two models scored on the same data. The grouping loss [30] is the single-model relative of our case-level part: what calibration leaves behind because inputs sharing a confidence level differ; ours contrasts two models within groups declared in advance. Transporting labels within a group is a within-group permutation of the input–label pairing, so one model’s case-level part is a conditional permutation importance [37] or model reliance [14] of the non-grouping part of its input, and the case-level part of a gain is the difference of the two models’ (Proposition 2). What the split adds is the exact additive accounting of a paired gain, including thresholded metrics, and paired inference for it.

Comparing two predictors. Inference on a paired score difference is the Diebold–Mariano test [12] and its conditional form [16]; at a single group our total-gain interval is a clustered Diebold–Mariano test (Corollary 3), so the

decomposition adds a split to what those tests already report. DeLong *et al.* [10] compare correlated ROC areas with a paired U-statistic, the same device we use for a group-conditional score gain. The case-level term is an order-two U-statistic [18], with a Hájek projection supplying its influence function [34].

3. Splitting a score gain

Setup. Let $q_0(\cdot | x)$ and $q_1(\cdot | x)$ be the predictive distributions of an old and a new frozen model over K labels, and $S(q, y)$ a bounded proper score, larger being better. Our default is the quadratic score $S(q, y) = 2q_y - \|q\|_2^2$, negative Brier loss up to a label-only constant; the log score is the second case we report. Let $C = \phi(X)$ be a discrete grouping both models observe, declared before the audit — for relation prediction, the ordered subject–object class pair. We call a value of C a *cell*. The gain vector

$$H_x(y) = S(q_1(\cdot | x), y) - S(q_0(\cdot | x), y), \quad y = 1, \dots, K, \quad (1)$$

can be evaluated at a counterfactual label without rerunning either model, and that is all the construction needs.

Transport. For a cell c , let $\Delta T_c = \mathbb{E}[H_X(Y) | C = c]$ be the observed gain and $\Delta P_c = \mathbb{E}_{H|c} \mathbb{E}_{Y'|c}[H_X(Y')]$ the gain when each case is scored against an *independent* label Y' drawn from the same cell, and set $\Delta R_c = \Delta T_c - \Delta P_c$. Transport keeps the cell and its label distribution and removes only the pairing between a case and its own label.

Theorem 1 (Transport and covariance identities). *For any integrable score contrast H ,*

$$\Delta T_c = \Delta P_c + \Delta R_c, \quad (2)$$

$$\Delta R_c = \sum_{y=1}^K \text{Cov}(H_X(y), \mathbb{1}\{Y = y\} | C = c). \quad (3)$$

For the quadratic score, with $\Delta q_y(X) = q_1(y | X) - q_0(y | X)$,

$$\Delta R_c = 2 \sum_{y=1}^K \text{Cov}(\Delta q_y(X), \mathbb{1}\{Y = y\} | C = c). \quad (3)$$

Consequently, if the gain vector is a function of c alone, $\Delta R_c = 0$.

Eq. (3) is what the split means. The new model earns case-level credit only insofar as its probability *change* rises on the label the individual case actually carries, relative to other cases in the same cell. Anything constant within a cell — shifting probability towards rare predicates for every (man, surfboard) relation alike — lands in ΔP , which is why we call it the group-level part. It is *not* simply “prior fit”: under

the quadratic score ΔP_c is the improvement in how close the model's mean forecast sits to the cell's label frequencies minus the increase in the dispersion of its forecasts (Proposition 1). The covariance form is due to Yates [45] and generalises: for every Bregman score, including the log score, ΔR_c is the same covariance with Δq_y replaced by the contrast of the y -th gradient coordinate (Theorem 2). All proofs are in the supplementary.

Why it matters which part a gain is in. In one cell with three cases labelled 1 and one labelled 2, against an old model predicting $(\frac{1}{2}, \frac{1}{2})$, a model predicting the cell's frequencies $(\frac{3}{4}, \frac{1}{4})$ for every case gains $+\frac{1}{8}$ of quadratic score, all group-level; a model equally far from the old one but pointed at each case's own label gains $+\frac{3}{8}$, all case-level. Shift the cell's labels to $(\frac{1}{4}, \frac{3}{4})$ and the first falls to $-\frac{3}{8}$, worse than the model it replaced, while the second stays at $+\frac{3}{8}$. In general, with $m_y(z) = \mathbb{E}[H_X(z) \mid Y = y, C = c]$, $\Delta R_c = \sum_{y < z} p(y)p(z) D(y, z)$ where $D(y, z) = m_y(y) - m_y(z) + m_z(z) - m_z(y)$: a label shift fixes every D and moves only the weights, so a case-level gain is less exposed to a shift than a group-level one, though with three or more labels it can still reverse (Appendix C).

Estimator. In a cell with $n_c \geq 2$ cases, write $H_i(y) = H_{x_i}(y)$ and estimate

$$\widehat{\Delta T}_c = \frac{1}{n_c} \sum_i H_i(y_i), \quad \widehat{\Delta P}_c = \frac{1}{n_c(n_c-1)} \sum_{i \neq j} H_i(y_j), \quad (4)$$

$$\widehat{\Delta R}_c = \binom{n_c}{2}^{-1} \sum_{i < j} A_{ij}, \quad (5)$$

$$A_{ij} = \frac{1}{2} \{H_i(y_i) + H_j(y_j) - H_i(y_j) - H_j(y_i)\}. \quad (6)$$

These satisfy $\widehat{\Delta T}_c = \widehat{\Delta P}_c + \widehat{\Delta R}_c$ exactly in every sample, and $\widehat{\Delta R}_c$ is an unbiased order-two U-statistic [18] for ΔR_c (Theorem 3). Transporting only to the *other* cases' labels matters: the all-case plug-in shrinks the case-level gain by $(n_c - 1)/n_c$ (Proposition 5). With m_{cy} the count of label y in cell c , case i 's transported score is

$$P_i = \frac{\sum_y m_{cy} H_i(y) - H_i(y_i)}{n_c - 1}, \quad (7)$$

so the whole estimator costs $O(NK)$ on cached predictions, with no fitted model, held-out fold or smoothing parameter. Cells are weighted by size; singletons cannot identify a within-cell counterfactual and are excluded and reported.

Inference. The Hájek influence contributions of the U-statistic (Appendix F), summed within each image, give one-way cluster-robust intervals, since relations from one photograph are not independent; the estimator is asymptotically normal under bounded scores and non-dominating

clusters (Theorem 4), and at a single cell the total-gain interval is a clustered Diebold–Mariano test [12] (Corollary 3).

Confidence matching. Because ΔR is a covariance between probability *movements* and labels, rescaling a model's confidence moves score between the two parts while changing no ranking (Sec. 4), so models that differ in calibration are not compared on raw outputs. We split the images at random, fit each model a temperature by maximum likelihood on one half, and split the gain on the other. A temperature cannot undo a class-wise shift of the operating point, so we also report matching by a temperature and a per-class bias, which preserves every within-cell ranking too.

Splitting mean recall. Nothing in Theorem 1 or the estimator uses properness: both hold for any contrast $H_x(y)$ that can be evaluated at a counterfactual label. Mean recall@ k is one. Let $h_x^{(m)}(y) \in \{0, 1\}$ say whether model m would recall relation x among its top k if its label were y ; replaying the benchmark's matching rules on cached scores gives it for every y (Appendix O). Pooling relations, mean recall is $\sum_i w(y_i) h_{x_i}(y_i)$ with $w(y) = 1/(Kn_y)$, so $H_x(y) = Nw(y)[h_x^{(1)}(y) - h_x^{(0)}(y)]$ makes $\widehat{\Delta T}$ its change; transport preserves every n_y , and restricting w to one tier or predicate splits that share exactly. One difference matters for reading the result. A hit is a thresholded decision, so moving a model's operating point — subtracting a log-prior from its logits, say — flips first the cases it already scored as most like the favoured label, and those carry that label more often: a case-level part with no new discrimination. We therefore read the split against an *operating-point control*, the old model's logits shifted by the training log-prior [28], and pair it with a threshold-free test, the *within-cell AUC*: the probability that a case labelled y has larger log-odds $\log q(y) - \log q(z)$ than one labelled z in the same cell. Averaged with weight $n_c(y)n_c(z)$, the number of comparisons, it weights a cell by about n_c^2 ; with weight $n_c(y)n_c(z)/n_c$ it combines cells as the split does. We report both. A temperature, and any change constant within a cell, leaves either unchanged.

Choosing the grouping. The split is conditional on ϕ . Merging cells changes it by an explicit between-cell covariance term of either sign (Proposition 3), so the finest grouping that the benchmark's construction defends is the conservative choice. A relevant variable left out of ϕ is the same term with the fully adjusted cell as the fine partition, and Proposition 4 bounds it by a single elicited correlation, in the spirit of the robustness value of Cinelli and Hazlett [7].

4. Does the split measure what it claims?

A released checkpoint cannot validate the split, because what the model can see is unknown. We validate it where the answer is known, first in simulation and then on Visual Genome predictors whose information content we control. All intervals are 95%.

Simulation. With 24 cells over five labels, an old model equal to the cell prior and a new one interpolating towards an oracle, the estimated case-level gain grows linearly with the injected signal; under a within-cell null the randomization test rejects in 0.030 of 200 datasets at level 0.05 (mean estimate -8.97×10^{-5}). For the interval we use on real data, we simulate relations clustered into images with a shared per-image effect and Zipf-distributed cell sizes (most cells hold two cases, most relations sit in a few large ones): the image-clustered interval covers in 94.0% of 500 runs, against 88.6% for an interval that treats relations as independent. With VG150’s own cell sizes it covers 96.3%; only when every cell holds two cases does it fall short, to 81.5% (Appendix J).

Three designs check that the two parts separate what they claim to. A *prior-only* improvement, where the new model corrects a noisy estimate of the cell prior, gives a case-level gain centred at zero (0.0002, sd 0.0035). An *unrecorded within-cell shortcut* — a binary feature varying inside each cell that the new model uses — is credited to the case-level part (+0.041, sd 0.004): the split measures all within-cell signal relative to the declared grouping, shortcut or not. And a *calibration-only* change, where the new model is a temperature-softened copy of an overconfident one carrying identical case-level information, is credited a large spurious case-level *loss* of -0.488 on raw outputs, which the confidence-matching protocol of Sec. 3 reduces to -0.0003 (sd 0.0017). That check is what matching is built to pass. A *class-wise shift of the operating point*, adding a fixed logit offset per label to the old model, leaves every within-cell ranking unchanged, yet survives temperature matching (+0.0106 for a logit-adjustment offset) and vanishes only when each model is matched by a temperature and a per-class bias (0.0000, sd 0.0003; Appendix Q.4). Both protocols are therefore part of the procedure.

Controlled predictors on VG150. We train predicate classifiers whose inputs we control on a VG150-style split we reconstruct from the released Visual Genome annotations, with the same 150-object and 50-predicate vocabulary and images divided 70/30 between fitting and audit (Appendix N.1); it is not the canonical split of Sec. 5. The predictors are: a frequency table that sees only the class pair (FREQ), an MLP on class embeddings (MLP-CLASS), and MLPs adding twelve box-geometry features (MLP-

SPATIAL, MLP-SPATIAL+). The audit covers 227,337 of 229,605 test relations (99.0%) in 6,346 class-pair cells.

MLP-CLASS beats FREQ by 0.03386 of quadratic score and the split assigns *all* of it to the group-level part: the case-level gain is 0.00000 with a zero-width interval. That zero is algebraic, not empirical — both models are functions of the class pair alone, so Theorem 1 forces it — and it is the sharpest check that nothing is credited to case-level evidence when none exists. Adding box geometry then yields a case-level gain of 0.01439 (CI [0.01373, 0.01504], $p = .002$) that no aggregate score isolates; against MLP-CLASS the total gain of 0.00885 even *understates* it, because the group-level part falls by 0.00554.

5. What TDE’s ten points bought

Tang *et al.* [39] release a causal MOTIFS PredCls checkpoint that yields two models from one set of weights: the baseline (MOTIFS-SUM), whose predicate logit sums a union-region visual term, a context term and a frequency prior indexed by the object pair, and TDE, which subtracts the same sum with an averaged context and so keeps only the context term minus its average. We ran it once with the authors’ code on the canonical VG150 test split, saving every branch’s logits, and replayed the official evaluator offline (Appendix O). The replay reproduces the official numbers to the fourth decimal: baseline $R@50 = 0.6612$ and $mR@50 = 0.1459$, TDE 0.4588 and 0.2476; without the graph constraint TDE’s mean recall *falls*, from 0.3260 to 0.2981. The audit covers 183,642 relations over 26,446 images in 7,127 cells; 98.9% fall in the 5,044 cells holding at least two.

An operating-point control. We add one model that needs no training: the baseline’s logits minus $\tau \log \pi(y)$, the training predicate prior, with $\tau = 1$ [28]. Within a cell this adds the same constant to every log-odds $\log q(y) - \log q(z)$, so it changes no within-cell ranking: whatever case-level part it shows is existing discrimination spent at a new operating point, not new discrimination. It reaches $mR@50 = 0.2343$ at $R@50 = 0.6437$.

Mean recall, split. Pooled over identified relations, mean recall@50 rises by 0.0972 (0.1017 on the official per-image average). Of that, +0.0841 is group-level and +0.0130 case-level, interval [+0.0093, +0.0168] (Tab. 1): 87% of the gain, interval [83%, 91%], survives shuffling every label among the relations that share its object pair. It lives in the body of the distribution. Splitting predicates by training frequency (9 head above 10,000 instances, 26 body, 15 tail below 500), the body gains +0.1224, of which +0.1082 is group-level; the head loses 0.0255; the tail gains 0.0003: of the 15 tail predicates the baseline recalls none and TDE one,

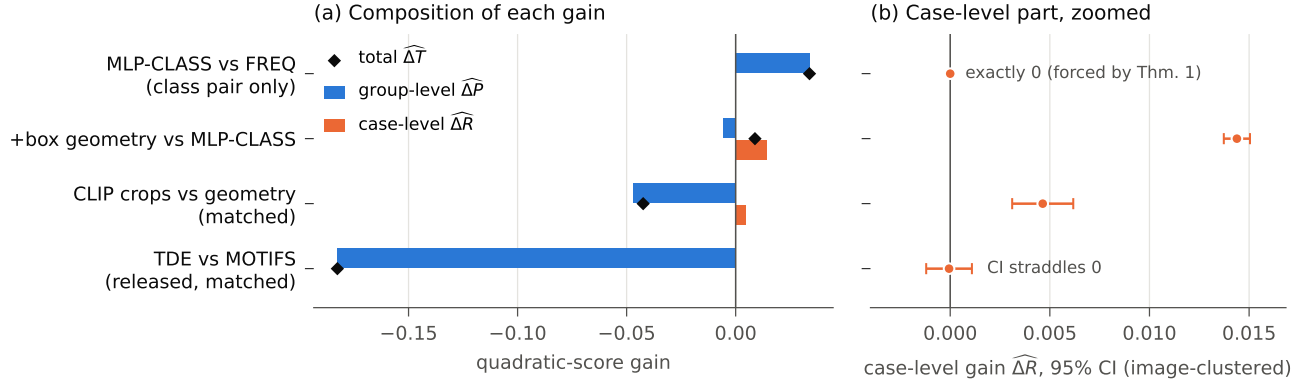


Figure 2. **Where each proper-score gain lives** (quadratic score). (a) Total gain $\widehat{\Delta T}$ (diamond) split exactly into group-level $\widehat{\Delta P}$ and case-level $\widehat{\Delta R}$. (b) $\widehat{\Delta R}$ with its image-clustered 95% interval. Class-pair-only models differ purely at group level (Theorem 1); geometry and CLIP crops buy case-level gain, CLIP despite losing overall; TDE’s change is group-level. CLIP and TDE rows are confidence-matched.

Table 1. **Mean recall@50 split** (graph constraint, micro-averaged over identified relations) against the MOTIFS baseline, except TDE–LA. LA: the logit-adjusted baseline; IETrans: the released IETrans+Rwt checkpoint, scored on the same relations against a separately trained baseline. Intervals 95%; case-level image-clustered. ΔAUC : change in within-cell AUC, weighted by comparison (c) or as the split weights cells (r); no calibration or group-level shift moves it (LA’s residue is float32 rounding). AUC intervals: image bootstrap.

	ΔmR	group	case [CI]	ΔAUC_c	ΔAUC_r
TDE	+0.0972	+0.0841	+0.0130	+0.0026	−0.0015
LA	+0.0941	+0.0802	+0.0139	+0.0001	+0.0003
TDE–LA	+0.0031	+0.0039	−0.0008	+0.0026	−0.0018
IETrans	+0.2204	+0.2143	+0.0060	−0.0093	−0.0235

to, at 0.016. Figure 3 shows the mechanism one predicate at a time. Recall of **parked on** rises from 0.000 to 0.886, almost all of it group-level: TDE now says **parked on** for (car, street). Recall of **on** falls by 0.558 in the same way.

The case-level part is matched by the control. The control reproduces the whole picture: +0.0941 of mean recall, +0.0802 group-level and +0.0139 case-level. Against it, TDE’s case-level part is −0.0008, interval [−0.0050, +0.0033]. So a case-level part of a thresholded metric is not by itself evidence of new discrimination. When the boundary between **on** and **parked on** moves, the cases that cross it are those the baseline already scored as more **parked on**-like, and they are more often labelled so; the covariance is real, but bought with discrimination the baseline had. The direct test is the within-cell AUC, which no prior shift or recalibration can move: TDE’s change is

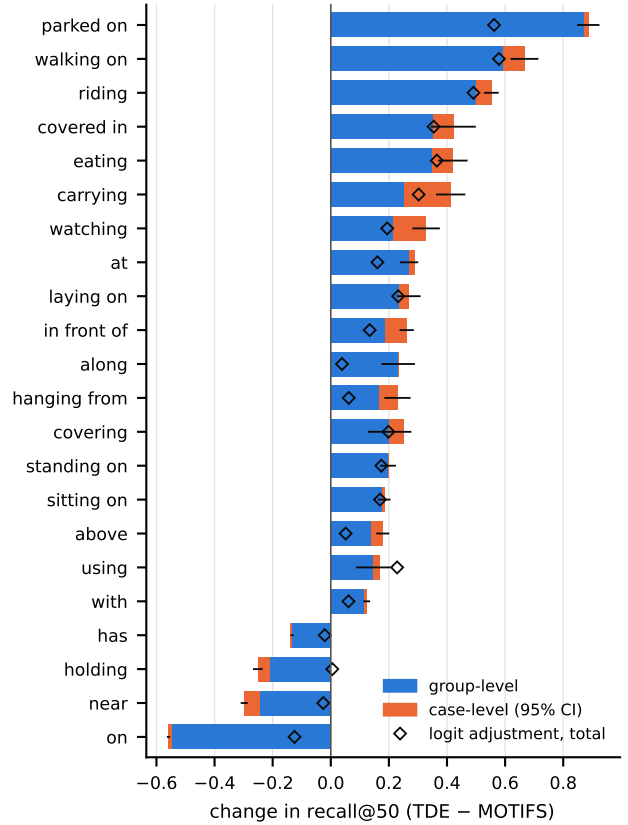


Figure 3. **Where TDE’s mean recall went**, for every predicate whose recall@50 moved by at least 0.12: group-level part (blue), case-level part (orange) with its image-clustered 95% interval, and the total under logit adjustment of the baseline (diamond), which changes no within-cell ranking.

+0.0026, interval [−0.0028, +0.0086], weighted by comparison, and −0.0015, [−0.0036, +0.0018], weighted as

the split weights cells. By predicate tier the comparison-weighted change is $+0.0068$, $[-0.0003, +0.0134]$, on the head-head pairs that carry 65% of its weight and -0.0051 , $[-0.0130, +0.0026]$, on the rest (Appendix Q.4): a pattern of redistribution, not a detected one.

The proper score depends on the protocol. With each model's temperature fitted on held-out images ($T^* = 2.50$ baseline, 0.92 TDE), TDE's quadratic-score change is -0.18264 , of which -0.18258 is group-level and -0.00006 , interval $[-0.00120, +0.00109]$, case-level (Fig. 2); over 20 calibration halves the case-level part stays within $[-0.00131, +0.00059]$. But a temperature cannot undo a class-wise shift of the operating point, and the control shows one: its case-level part is $+0.0039$. Matching a temperature and a per-class bias for each model, which also preserves every within-cell ranking, makes the control identical to the baseline and TDE's case-level part $+0.00445$, interval $[+0.00354, +0.00536]$; one temperature shared by both models makes it -0.0184 ; the log score gives $+0.04396$ (Appendices Q.1 and Q.4). The sign of TDE's case-level part depends on the protocol. Merging synonymous predicates (32 classes) or collapsing them into three families changes none of this (Appendix Q.2).

What that licenses. On this checkpoint the ten points of mean recall are a change in which predicates the model favours within each object pair. A one-line prior shift that ranks the relations of every cell exactly as the baseline did reproduces the gain, its split and its tier profile at under a tenth of TDE's cost in $R@50$ (0.0175 against 0.2023), and a better $F@50$, the harmonic mean of $R@50$ and $mR@50$ (0.344 against 0.322). That a prior correction rivals debiasing is known [6, 41]; the split shows which part of TDE's gain it reproduces. No measure we tried detects a change in within-cell ranking, and the proper-score case-level part, small under every protocol, has no protocol-free sign.

Why. Under the released SUM fusion, TDE's change from the baseline is three subtractions, and splitting along them answers why; the parts add up to TDE's exactly (Appendix O.1). Removing the frequency prior, constant within every cell, moves mean recall by -0.0009 and the within-cell AUC not at all. Removing the visual (union-region) term adds $+0.0164$, with a case-level part of $+0.0016$, interval $[-0.0006, +0.0037]$: within a cell, the context term already knew what the visual term did. Subtracting the averaged context supplies $+0.0817$ of the 0.0972. That term is, to 99.96% of its energy, one vector shared by every relation in every image, correlated $+0.75$ with the log training prior. In PredCls, TDE's counterfactual is to first order a learned logit adjustment of the context branch, which is why the explicit one reproduces it.

A second method. IETrans [49] relabels training data, moving general predicates to specific ones within the same object pair, and adds the training log-prior at test time. Its released Neural-Motifs PredCls checkpoint is the IETrans+Rwt variant, trained with a reweighted loss as well; we ran it once with the authors' evaluation command and re-scored it the same way (Appendix O.2). It reaches $mR@50 = 0.3587$ (0.3576 on the authors' test set, which drops duplicate relations): $+0.2204$ over the baseline, of which $+0.2143$ is group-level and $+0.0060$ case-level, interval $[-0.0031, +0.0151]$. Unlike TDE it reaches the tail ($+0.0889$), again group-level. Its within-cell discrimination falls under every measure: matched case-level parts -0.0162 (quadratic) and -0.0544 (log), -0.0168 with a per-class bias as well, and the AUC change weighted as the split weights cells is -0.0235 , interval $[-0.0275, -0.0182]$, concentrated on head-body and body-body pairs. Both methods' gains are group-level; with TDE no change in within-cell ranking is detected, while IETrans+Rwt loses some. No IETrans baseline was released, so this compares two separately trained models and does not isolate the effect of relabelling.

Beyond PredCls. The same release holds SGCLs and SGDet checkpoints, where object classes (and in SGDet boxes) are predicted. One TDE pass still yields all three models, which share every object prediction, so the ground-truth pair remains a common grouping; we replay the evaluator inside the inference loop and reproduce the released recalls (Appendix Q.5). In SGCLs TDE raises $mR@50$ by $+0.0517$, 99% of it group-level; the control gains $+0.0518$ at $R@50 = 0.384$ against TDE's 0.263; and TDE's within-cell ranking gets worse, by -0.0128 of relation-weighted AUC, interval $[-0.0171, -0.0074]$. In SGDet, on the full test set, the control again gains more than TDE ($+0.0418$ against $+0.0310$) at nearly twice its $R@50$.

6. The model every metric ranks last

The predictors of Sec. 4 never see an image: they read the class pair and, at most, box coordinates from the annotation. On the same reconstructed split we add one that reads pixels — subject and object regions cropped from the image and encoded by a frozen CLIP ViT-B/32 [31], concatenated with the class embeddings MLP-CLASS uses. CLIP is not fine-tuned, so any case-level gain reflects what pretraining had already recovered. All three compared models are retrained on one fixed subsample of 100,000 training relations, so differences isolate the features rather than the amount of data.

By every aggregate measure the CLIP model is the worst of the three: top-1 accuracy 0.6156 against 0.6467 for the geometry model, and a quadratic score 0.04642 lower. An evaluator reading either number would discard it. The split

disagrees. With each model’s confidence matched on held-out images, the CLIP model loses 0.04705 of group-level score against the geometry model while *gaining* 0.00464 of case-level score (CI [+0.00311, +0.00618]); on raw outputs the figures are 0.05282 and 0.00640 (CI [0.00513, 0.00767], $p = .002$). Its whole deficit is group-level (Fig. 2). Against the class-only model, pixels buy 0.01663 of case-level score on raw outputs where box geometry buys 0.01023, with well-separated intervals.

A larger case-level part is not the same as better ranking, and here the two disagree. With each model’s prior matched within cells, the CLIP model still ranks the correct predicate first less often than the geometry model: top-1 accuracy 2.17 points lower, MRR 1.67 lower, recall@5 1.07 lower. So the CLIP model carries more case-level covariance without being the better ranker. The within-cell AUC, which no calibration or prior shift can move, does not favour it either: weighted by comparison, 0.5408 against 0.5323, a difference of +0.0085 with interval [−0.0062, +0.0214]; weighted as the split weights cells, geometry ranks better, −0.0067, [−0.0120, −0.0019]. Both models clear the class-only model’s 0.5 (by +0.0408 and +0.0323, comparison-weighted). The CLIP model’s case-level advantage is real covariance, not better ranking.

Seeds. Over five training seeds (seed 0 reproduces the predictions above exactly; Appendix P), the CLIP model’s matched case-level advantage over geometry stays between +0.00413 and +0.00474, every interval excluding zero, its accuracy stays below geometry’s, and its comparison-weighted AUC difference is positive in all five but excludes zero in one. Adding the CLIP features to geometry, rather than replacing it, doubles that advantage (+0.00807 to +0.00949).

Under a label shift. A label shift moves group-level parts and only reweights case-level ones (Sec. 3), so the split says which parts of a gain a shift can reverse. We reweight the test relations along a path from the benchmark’s predicate mix to uniform frequencies within each object pair (Appendix Q.3). At the uniform end, seven of the eight group-level parts of Sec. 5 and this section change sign, and the totals follow: the CLIP model now beats geometry by +0.02981. Five of the six case-level parts that excluded zero keep their sign and grow. The exceptions are TDE’s: its part against the baseline rises steadily along the path, from −0.00006 to +0.01430, interval [+0.01116, +0.01744], and against the control from −0.00398 to +0.00723, while the AUC separates it from the baseline at neither end. TDE gains discrimination between some predicate pairs and loses it between others, and the benchmark’s mix weights the two to zero.

Beyond scene graphs. Nothing in Sec. 3 is specific to relations; the supplementary applies the estimator to long-tailed image classification (CIFAR-100-LT, across seeds and imbalance ratios) and to long-tailed text classification. The constraint on the grouping bites there: the fine class label cannot serve as a cell, because it forces the case-level part to zero by construction, an artefact that reads like a null.

7. Limitations and conclusion

Limitations. Five bound what the split can say. *It is conditional on the grouping.* The case-level part measures discrimination among relations sharing an object pair; two equally fine partitions of the same cases can disagree completely about whether a gain is case-level (Appendix K), so declaring ϕ is a modelling decision that the benchmark, not the submitting author, should make before a leaderboard opens. *It credits any within-cell signal,* including shortcuts nobody recorded, and our own bound on that exposure is not reassuring: for the controlled-predictor pair of Sec. 4, an unrecorded covariate correlated at 0.061 with both parts would account for the entire case-level gain (Appendix I). *It is not invariant to recalibration, to the operating point or to the label mix:* a temperature does not undo a class-wise shift of the operating point (Sec. 4), so we report matching with a temperature and with a per-class bias as well; the within-cell AUC, invariant to both, gives up exact additivity; and a case-level null can be a cancellation that another label mix undoes (Sec. 6). *Its interval assumes cells that are not tiny:* with every cell holding two cases it covers 81.5%, not 95%, and from three cases up it reaches nominal coverage (Appendix J); 1.0% of VG150’s audited relations sit in two-case cells. And *it needs scores for every label:* most benchmarks archive only ranked top- k predictions, and recovering the rest took a rerun of the released checkpoint: that, not compute, blocks routine use.

Conclusion. A ten-point mean-recall gain is two achievements summed, and the field’s metrics report the sum. Scoring each prediction against another case’s label from the same group separates them exactly. On two released debiasing methods the gains are almost entirely group-level: TDE’s is matched by a ranking-preserving logit adjustment and leaves within-group ranking unchanged as far as we can detect; IETrans’s costs within-group discrimination. We suggest that benchmarks whose labels lean on a feature every model sees report, beside the headline metric, the declared grouping with its coverage and cell sizes, both signed parts with intervals under both matching protocols, the within-group AUC under both weightings, the parts along a path of label mixes, and, for a thresholded metric, the operating-point control; each model’s within-group reliance (Proposition 2) is a one-column summary a leaderboard can carry. Code and results are in the supplement.

References

- [1] Aishwarya Agrawal, Dhruv Batra, Devi Parikh, and Anirudha Kembhavi. Don't just assume; look and answer: Overcoming priors for visual question answering. In *CVPR*, pages 4971–4980, 2018. 3
- [2] Jochen Bröcker. Reliability, sufficiency, and the decomposition of proper scores. *Quarterly Journal of the Royal Meteorological Society*, 135(643):1512–1519, 2009. 3
- [3] Kaidi Cao, Colin Wei, Adrien Gaidon, Nikos Arechiga, and Tengyu Ma. Learning imbalanced datasets with label-distribution-aware margin loss. In *NeurIPS*, 2019. 3, 11, 19
- [4] Chao Chen, Yibing Zhan, Baosheng Yu, Liu Liu, Yong Luo, and Bo Du. Resistance training using prior bias: Toward unbiased scene graph generation. In *AAAI*, pages 212–220, 2022. 3
- [5] Tianshui Chen, Weihao Yu, Riquan Chen, and Liang Lin. Knowledge-embedded routing network for scene graph generation. In *CVPR*, pages 6156–6164, 2019. 2
- [6] Meng-Jiun Chiou, Henghui Ding, Hanshu Yan, Changhu Wang, Roger Zimmermann, and Jiashi Feng. Recovering the unbiased scene graphs from the biased ones. In *ACM International Conference on Multimedia*, pages 1581–1590, 2021. 3, 7
- [7] Carlos Cinelli and Chad Hazlett. Making sense of sensitivity: Extending omitted variable bias. *Journal of the Royal Statistical Society: Series B*, 82(1):39–67, 2020. 4, 13
- [8] Yin Cui, Menglin Jia, Tsung-Yi Lin, Yang Song, and Serge Belongie. Class-balanced loss based on effective number of samples. In *CVPR*, pages 9268–9277, 2019. 11, 19
- [9] Morris H. DeGroot and Stephen E. Fienberg. The comparison and evaluation of forecasters. *Journal of the Royal Statistical Society: Series D (The Statistician)*, 32(1-2):12–22, 1983. 3
- [10] Elizabeth R. DeLong, David M. DeLong, and Daniel L. Clarke-Pearson. Comparing the areas under two or more correlated receiver operating characteristic curves: A non-parametric approach. *Biometrics*, 44(3):837–845, 1988. 3, 13
- [11] Alakh Desai, Tz-Ying Wu, Subarna Tripathi, and Nuno Vasconcelos. Learning of visual relations: The devil is in the tails. In *ICCV*, pages 15384–15393, 2021. 3
- [12] Francis X Diebold and Roberto S Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995. 3, 4
- [13] Xingning Dong, Tian Gan, Xueming Song, Jianlong Wu, Yuan Cheng, and Liqiang Nie. Stacked hybrid-attention and group collaborative learning for unbiased scene graph generation. In *CVPR*, pages 19405–19414, 2022. 3
- [14] Aaron Fisher, Cynthia Rudin, and Francesca Dominici. All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously. *Journal of Machine Learning Research*, 20(177):1–81, 2019. 3, 2
- [15] Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673, 2020. 3
- [16] Raffaella Giacomini and Halbert White. Tests of conditional predictive ability. *Econometrica*, 74(6):1545–1578, 2006. 3
- [17] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, pages 770–778, 2016. 19
- [18] Wassily Hoeffding. A class of statistics with asymptotically normal distribution. *The Annals of Mathematical Statistics*, 19(3):293–325, 1948. 3, 4, 7
- [19] Corentin Kervadec, Grigory Antipov, Moez Baccouche, and Christian Wolf. Roses are red, violets are blue... but should vqa expect them to? In *CVPR*, pages 2775–2784, 2021. 2, 3
- [20] Ranjay Krishna, Yuke Zhu, Oliver Groth, Justin Johnson, Kenji Hata, Joshua Kravitz, Stephanie Chen, Yannis Kalantidis, Li-Jia Li, David A Shamma, et al. Visual genome: Connecting language and vision using crowdsourced dense image annotations. *International Journal of Computer Vision*, 123(1):32–73, 2017. 2, 14
- [21] Ken Lang. Newswelder: Learning to filter netnews. In *ICML*, pages 331–339, 1995. 22
- [22] Lin Li, Long Chen, Yifeng Huang, Zhimeng Zhang, Songyang Zhang, and Jun Xiao. The devil is in the labels: Noisy label correction for robust scene graph generation. In *CVPR*, pages 18847–18856, 2022. 3
- [23] Rongjie Li, Songyang Zhang, Bo Wan, and Xuming He. Bipartite graph network with adaptive message passing for unbiased scene graph generation. In *CVPR*, pages 11104–11114, 2021. 2, 3
- [24] Rongjie Li, Songyang Zhang, Dahua Lin, Kai Chen, and Xuming He. From pixels to graphs: Open-vocabulary scene graph generation with vision-language models. In *CVPR*, 2024. 3
- [25] Xingchen Li, Long Chen, Jian Shao, Shaoning Xiao, Songyang Zhang, and Jun Xiao. Rethinking the evaluation of unbiased scene graph generation. In *BMVC*, 2022. 2, 3
- [26] Zachary C Lipton, Yu-Xiang Wang, and Alexander Smola. Detecting and correcting for label shift with black box predictors. In *ICML*, pages 3122–3130, 2018. 4
- [27] Cewu Lu, Ranjay Krishna, Michael Bernstein, and Li Fei-Fei. Visual relationship detection with language priors. In *ECCV*, pages 852–869, 2016. 2
- [28] Aditya Krishna Menon, Sadeep Jayasumana, Ankit Singh Rawat, Himanshu Jain, Andreas Veit, and Sanjiv Kumar. Long-tail learning via logit adjustment. In *ICLR*, 2021. 2, 3, 4, 5, 20
- [29] Allan H. Murphy. A new vector partition of the probability score. *Journal of Applied Meteorology*, 12(4):595–600, 1973. 3
- [30] Alexandre Perez-Lebel, Marine Le Morvan, and Gaël Varoquaux. Beyond calibration: Estimating the grouping loss of modern neural networks. In *International Conference on Learning Representations*, 2023. 3
- [31] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, 680
681
682
683
684
685
686
687
688
689
690
691
692
693
694
695
696
697
698
699
700
701
702
703
704
705
706
707
708
709
710
711
712
713
714
715
716
717
718
719
720
721
722
723
724
725
726
727
728
729
730
731
732
733
734
735
736

- 737 Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen
738 Krueger, and Ilya Sutskever. Learning transferable visual
739 models from natural language supervision. In *ICML*, pages
740 8748–8763, 2021. 2, 7, 16, 34
- 741 [32] Paul R. Rosenbaum. *Observational Studies*. Springer, New
742 York, 2 edition, 2002. 5, 13
- 743 [33] Marco Saerens, Patrice Latinne, and Christine Decaestecker.
744 Adjusting the outputs of a classifier to new a priori probabili-
745 ties: A simple procedure. *Neural Computation*, 14(1):21–41,
746 2002. 4
- 747 [34] Robert J. Serfling. *Approximation Theorems of Mathemati-
748 cal Statistics*. John Wiley & Sons, New York, 1980. 3, 9,
749 12
- 750 [35] Robik Shrestha, Kushal Kafle, and Christopher Kanan. A
751 negative case analysis of visual grounding methods for VQA.
752 In *Proceedings of the 58th Annual Meeting of the Association
753 for Computational Linguistics*, pages 8172–8181, 2020. 3
- 754 [36] Stefan Siegert. Variance estimation for brier score decompo-
755 sition. *Quarterly Journal of the Royal Meteorological Soci-
756 ety*, 140(682):1771–1777, 2014. 3
- 757 [37] Carolin Strobl, Anne-Laure Boulesteix, Thomas Kneib,
758 Thomas Augustin, and Achim Zeileis. Conditional variable
759 importance for random forests. *BMC Bioinformatics*, 9:307,
760 2008. 3, 2
- 761 [38] Kaihua Tang, Hanwang Zhang, Baoyuan Wu, Wenhan Luo,
762 and Wei Liu. Learning to compose dynamic tree structures
763 for visual contexts. In *CVPR*, pages 6612–6621, 2019. 2
- 764 [39] Kaihua Tang, Yulei Niu, Jianqiang Huang, Jiaxin Shi, and
765 Hanwang Zhang. Unbiased scene graph generation from bi-
766 ased training. In *CVPR*, pages 3716–3725, 2020. 2, 5
- 767 [40] Damien Teney, Ehsan Abbasnejad, Kushal Kafle, Robik
768 Shrestha, Christopher Kanan, and Anton van den Hengel.
769 On the value of out-of-distribution testing: An example of
770 Goodhart’s law. In *Advances in Neural Information Process-
771 ing Systems*, 2020. 3
- 772 [41] Yao Teng and Limin Wang. Structured sparse R-CNN for di-
773 rect scene graph generation. In *CVPR*, pages 19415–19424,
774 2022. 3, 7
- 775 [42] Aad W. van der Vaart. *Asymptotic Statistics*. Cambridge
776 University Press, Cambridge, UK, 1998. 9, 12
- 777 [43] Danfei Xu, Yuke Zhu, Christopher B Choy, and Li Fei-Fei.
778 Scene graph generation by iterative message passing. In
779 *CVPR*, pages 3097–3106, 2017. 2, 14
- 780 [44] Shaotian Yan, Chen Shen, Zhongming Jin, Jianqiang Huang,
781 Rongxin Jiang, Yaowu Chen, and Xian-Sheng Hua. PCPL:
782 Predicate-correlation perception learning for unbiased scene
783 graph generation. In *ACM International Conference on Mul-
784 timedia*, pages 265–273, 2020. 3
- 785 [45] J. Frank Yates. External correspondence: Decompositions
786 of the mean probability score. *Organizational Behavior and
787 Human Performance*, 30(1):132–156, 1982. 3, 4, 14
- 788 [46] Jing Yu, Yuan Chai, Yujing Wang, Yue Hu, and Qi Wu.
789 CogTree: Cognition tree loss for unbiased scene graph gen-
790 eration. In *IJCAI*, pages 1274–1280, 2021. 3
- 791 [47] Mert Yuksekgonul, Federico Bianchi, Pratyusha Kalluri,
792 Dan Jurafsky, and James Zou. When and why vision-
793 language models behave like bags-of-words, and what to do
about it? In *International Conference on Learning Repre-
sentations*, 2023. 34
- [48] Rowan Zellers, Mark Yatskar, Sam Thomson, and Yejin
Choi. Neural motifs: Scene graph parsing with global con-
text. In *CVPR*, pages 5831–5840, 2018. 2
- [49] Ao Zhang, Yuan Yao, Qianyu Chen, Wei Ji, Zhiyuan Liu,
Maosong Sun, and Tat-Seng Chua. Fine-grained scene graph
generation with data transfer. In *Computer Vision – ECCV
2022*, pages 409–424, 2022. 2, 3, 7
- [50] Chaofan Zheng, Xinyu Lyu, Lianli Gao, Bo Dai, and
Jingkuan Song. Prototype-based embedding network for
scene graph generation. In *CVPR*, pages 22783–22792,
2023. 3
- [51] Wenhao Zhou, Faqiang Liu, Hao Zheng, and Rong Zhao.
Mitigating data bias and ensuring reliable evaluation of ai
models with shortcut hull learning. *Nature Communications*,
16, 2025. 3